Groundwater Recharge in the Talvera Basin

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1. Introduction. The Talvera basin supplies drinking water to about 180,000 inhabitants. Its aquifer is recharged mainly in winter, and the dry summers of the last decade raised the question of how much water can be pumped without lowering the water table. The working group reviewed the study design at each of its monthly meetings and recorded every decision in the minutes, so that the reasoning behind each choice can be traced by later teams.

2. Study area and data. The basin covers 412 square kilometres between the Serra hills and the river Talvera. The working group reviewed the data sources at each of its monthly meetings and recorded every decision in the minutes, so that the reasoning behind each choice can be traced by later teams.

2.1. Monitoring wells. Water levels were recorded every hour in 37 monitoring wells from January 2019 to December 2025. Six wells with gaps longer than three months were excluded, which left 31 wells for the analysis.

2.2. Rainfall records. Daily rainfall comes from the eight stations of the regional weather service. The mean annual rainfall over the period was 640 millimetres, with a minimum of 410 millimetres in 2022.

3. Recharge estimation. Recharge was estimated with the water table fluctuation method, which multiplies the rise of the water table after each rain event by the specific yield of the aquifer. The working group reviewed the estimation method at each of its monthly meetings and recorded every decision in the minutes, so that the reasoning behind each choice can be traced by later teams.

Remark 1. The specific yield was set to 0.08 for the sandy layers and 0.03 for the clay layers, the values measured in the pumping tests of 2018.

3.1. Uncertainty. A Monte Carlo analysis with 5,000 draws of the specific yield gave a 90 percent interval of 52 to 81 millimetres per year for the mean recharge.

4. Results. The mean annual recharge was 66 millimetres, that is about 27 million cubic metres per year over the basin. Recharge fell to 19 millimetres in the dry year 2022.

5. Discussion. Pumping currently takes 24 million cubic metres per year, close to the mean recharge. The authors recommend capping the pumping at 20 million cubic metres in dry years and building two infiltration basins near the river.

References

[1] Healy, R. Estimating groundwater recharge. Fictional Press, 2010.